Reg. No. | | | | | | | | | | |

[18MEDE748]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

ALTERNATE FUELS AND RENEWABLE ENERGY

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Explain the requirements of the alternate fuels for the IC engines.	**06**
	b.	List and elaborate the important properties of alternate fuels in an IC engines.	**08**
2.	**a.**	Illustrate the working principle of LPG conversion kit fitted on gasoline engine with a neat diagram.	**10**
	b.	Identify the merits and demerits of Biogas.	**04**
3.	**a.**	List and elaborate the important properties required of biodiesel be used as an alternate fuels.	**06**
	b.	Illustrate the process of biodiesel transesterification process and discuss the importance of alcohol and catalyst usage in this process.	**08**
		OR	
4.	**a.**	State the advantages and disadvantages of Biodiesel over pure diesel to	**04**
	b.	Enumerate and brief the common emissions produced from the combustion of biodiesel.	**10**
5.	**a.**	Write short notes on solar radiation.	**04**
	b.	Describe the main elements of solar photovoltaic cells with neat sketch.	**10**
		OR	
6.	**a.**	State the advantages and disadvantages of solar powered vehicle over conventional vehicles.	**06**
	b.	Explain the working of solar powered vehicle stating its layout and components.	**08**
7.	**a.**	What are the different types of batteries used in the electric vehicles? Explain each of them.	**06**
	b.	What are the advantages and limitations of: i. High energy batteries ii. Power density batteries.	**08**
		OR	
8.	**a.**	Explain the control system of electric and hybrid electric vehicles.	**06**
	b.	Explain the layout of the hybrid vehicle with a neat sketch.	**08**